A-A-53802
27 March 1989

COMMERCIAL ITEM DESCRIPTION

PLUGGERS, PULP CANAL, ENDODONTIC

The General Services Administration has authorized the use of this Commercial Item Description.

Salient characteristics:

This commercial item description covers double-ended pulp canal pluggers for use in endodontic procedures.

Types and numbers. Pluggers shall be of the following types and plugger numbers as specified (See Notes).

Type I - One piece.

Plugger No. 1-3
Plugger No. 5-7
Plugger No. 9-11

Type II - Three piece.

Plugger No. 1-3
Plugger No. 5-7
Plugger No. 9-11

Material. The pluggers shall be fabricated from the stainless steel specified below and shall be in accordance with ASTM F899. Chemical analysis shall be tested as specified herein.

Materials.

Type I (One piece pluggers)	Class 4, type 440A.
Type II (Three piece pluggers)	
Operative ends	Class 5, type 631.
Handle A, B and C	Class 3, type 302, 303, 304, or Class 4, 416, or class 6, type 430F.
Handle D	Chromium over nickel plated brass.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Hardness. Materials shall be properly stress-relieved and passivated. After stress relieving and passivation, the Rockwell hardness of the pluggers shall be as specified below when tested as specified herein.

Hardness.

Type I (One piece pluggers)	
Operative ends	C50 to C56
Shank	C40 to C50
Type II (Three piece pluggers)	
Operative ends and shank	C40 to C50

Design and dimensions. The pluggers shall be double-ended endodontic pluggers suitable for use in plugging and condensing gutta percha points in root canals. The pluggers shall be of the type and number as specified herein. The design and dimensions of the pluggers shall conform to the Defense Personnel Support Center (DPSC) Drawing 24329 for the applicable type and plugger number. Pluggers shall be furnished either as Type I or Type II instrument for the applicable plugger number. Type II (three piece) pluggers shall be furnished with Handle A, B, C or D conforming to the drawing.

Construction. Type I pluggers shall be one piece, double-ended pluggers. Type II pluggers shall be three piece, double-ended pluggers.

One piece instruments. The handles of one piece (Type I) pluggers shall be octagonal in cross-section and in accordance with the specified drawing. Handles shall be straight.

Three piece instruments. The handles of three piece (Type II) pluggers shall be one piece, hollow, round handles. The finger grip portion of the handle shall be finely serrated or fine diamond knurled. The operative ends shall be firmly and permanently secured to the handle. The junction between the operative end and handle shall be smoothly finished and free of crevices.

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Deflection. The operative ends of Type I and Type II pluggers shall show no more than 1/4 degree permanent deflection following a bend of 8 degrees minimum for plugger numbers 1-3 and 5-7 (4 degrees minimum for numbers 9-11); no more than 2 degrees permanent deflection following a bend of 10 degrees minimum for numbers 1-3 and 5-7 (5 degrees minimum for numbers 9-11); and no sign of rupture or fracture following a bend of 20 degrees minimum for numbers 1-3 and 5-7 (10 degrees minimum for numbers 9-11), when tested as specified herein.

Ductility. Pluggers (operative ends) manufactured from class 5 (Type 631) shall withstand a minimum of two 90 degree cold bend, with no signs of rupture or fracture, when tested as specified herein.

Finish.

Operative ends and shanks. Operative ends and shanks shall have a bright or mirror finish and shall be free from defects such as coarse grind marks, cracks, burrs, or other defects when viewed as specified herein. The pluggers shall be uniformly and symmetrically tapered, and smoothly finished (including the flat face on the tip of the operative end).

Handles. Handles of one or three piece instruments (except for the finely serrated or fine knurled finger grip portion) shall have a mirror or bright polished finish. Serrated or knurled portions shall be free of barbs and shall provide adequate finger grip during use. Serrations or knurling shall be uniform and complete.

Corrosion resistance. One piece instruments or handles fabricated from martensitic stainless steel (Class 4, Type 440A or 416) or ferritic stainless steel (AISI Type 430F) shall meet the copper sulfate requirements of ASTM F1089, when tested as specified herein. The complete three piece instrument shall meet the boil test requirements of ASTM F1089, when tested as specified herein.

Identification marking. Pluggers shall be permanently and legibly marked with applicable plugger number. All marking shall be on the smooth face of Type I plugger handles (one piece plugger, octagonal handle) and on the smooth portion of Type II plugger handles (three piece, round handle). Each plugger shall be marked with the name or registered trademark of the manufacturer (and contractor, if other than the manufacturer) and the word "Stainless" when applicable.

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In addition, instruments of foreign (other than U.S.) manufacture shall also be marked with the name of the country of origin. As an alternate to the word "Stainless" the use of a registered trade name identifying the steel as stainless shall be acceptable, provided prior approval is received from the contracting officer.

Workmanship. The pluggers shall be free from cracks, crevices, pits, burrs and other defects which detract from their appearance or impair their serviceability, proper functioning, and intended use.

Unit. Each (EA). One plugger of one type and number, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5 (percent defective) and an inspection level of II. Examination shall include, but shall not be limited to the defects listed below. The unit of product for examination purposes shall be one plugger.

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Defects

Design not as specified for the applicable type and number.
Construction not as specified for the applicable type and number.
Finish of operative ends not as specified *
Operative ends not free of coarse grind marks, cracks, burrs or other imperfections.
Operative ends bent or distorted.
Operative ends not uniformly and symmetrically tapered.
Handle finish not as specified.
Identification marking not present, not properly located, illegible, or incomplete.
Finish of handle not free of coarse grind marks.
Plugger not free from dirt, oil, compound, or foreign matter.
Handle not straight.
Handle serrations or knurling not as specified.
Handle serrations or knurling either not uniform or not complete.

* Examination under magnification required.

Examination under magnification. The instruments shall be examined under four power (4X) magnification when specified. Examination under magnification shall include, but shall not be limited to, the defects listed herein.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as specified below, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one plugger.

Test criteria

Component	Characteristics	Inspection level
Entire instrument	Chemical analysis	**
Operative ends and shank	Hardness	S-2
Entire instrument	Dimensions	S-2
Operative Ends (Type I and Type II)	Deflection	S-2
Operative Ends (Class 5 Material)	Ductility	S-2
Entire instrument or handles	Copper sulfate	S-2
Entire instrument	Boil test	S-2

**A sufficient quantity of each lot of raw steel used in the fabrication of the instrument shall be selected for chemical analysis.

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Tests. Tests shall be conducted to determine compliance with the requirements of this specification. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Chemical analysis. In the absence of certificates of quality, the chemical analysis of the instruments shall be determined in accordance with methods 111 or 112 of FED-STD-151. In the event of a conflict, method 111 shall govern.

Hardness. Hardness shall be determined in accordance with ASTM E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

Dimensions test. Tests of dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with them.

Deflection test. The deflection test shall be conducted with the first 5/32 inch of the operative end (distal end) of the plugger clamped by means of a hemostatic forcep, vise or other suitable device. The tip shall be bent back (away from the handle in the direction the tip would deflect during normal usage) the amount specified previously for the applicable plugger and then examined for the criteria previously specified after release. Examination for rupture or fracture shall be conducted under a minimum of 30 power (30X) magnification. Tests shall be conducted at room temperature.

Ductility test. The first 1/4 inch of the operative end (distal end) of the plugger shall be clamped in a vise or other suitable device, the jaws of which shall be rounded to a radius of 1/16 inch. The plugger operative end shall be bent back and forth over this radius at a uniform rate. Each 90 degree of travel shall be considered one bend. Examination for rupture or fracture shall be conducted under a minimum of 30 power (30X) magnification. Tests shall be conducted at room temperature.

Copper sulfate test. The copper sulfate test shall be conducted in accordance with ASTM F1089.

Boil test. The boil test shall be conducted in accordance with ASTM F1089.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification	Type
6520-00-083-6577	PLUGGER, Pulp Canal, Endodontic, No. 1-2: Double-ended, contra-angle point	I, II

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6520-00-083-6579	PLUGGER, Pulp Canal, Endodontic, I, II No. 5-7: Double-ended, contra-angle point
6520-00-083-6582	PLUGGER, Pulp Canal, Endodontic, I, II No. 9-11: Double-ended, contra-angle point

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-1898

LOCATION: NCR Disk No. 7

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